Educational Data Mining Student Performance Research

Paper 1

https://d.docs.live.net/ec55ae27f4c6f109/Desktop/EDM%20PROJECT/Paper1.pdf

Key Takeaways (For our Research)

- Used 450 students' grade data (65 courses, 9 semesters).
- Applied 5 ML algorithms: kNN, Random Forest, Decision Tree, Logistic Regression, Neural Network.
- Binary classification (Pass/Fail) performed much better than multi-class prediction.
- Neural Networks, kNN, and Random Forest gave the highest F1 scores (best ≈ 0.93).
- Used correlation-based feature selection — courses with correlation ≥ 0.3 improved prediction accuracy.
- Adding too many weakly correlated courses reduced performance.
- Used F1-score because dataset was imbalanced.
- Only academic grades were used (no LMS activity or behavioral data).

What's Useful for us

- Use binary + multi-class comparison in your paper.
- Apply correlation-based feature selection.
- Use F1-score instead of only accuracy.
- Improve by adding behavioral/LMS data (gap in this paper).

Paper 2

https://d.docs.live.net/ec55ae27f4c6f109/Desktop/EDM%20PROJECT/Paper2.pdf

Key Takeaways (For our Research)

- Introduces **Federated Learning (FL)** for privacy-preserving Educational Data Mining.
- Instead of sharing student data, institutions share **model updates only**.
- Solves privacy issues related to sensitive data (grades, behavior, demographics).
- Handles multi-institution collaboration without centralized data storage.
- Highlights **non-IID data problem** (different institutions have different data distributions).
- Discusses privacy techniques:
    - Differential Privacy
    - Secure aggregation
    - Encryption
- Mentions trade-off between **privacy and model accuracy**.
- Suggests FL for:
    - Student performance prediction
    - Dropout detection
    - Personalized learning systems

What's Useful for us

- You can mention FL in **literature review or future work**.
- Add discussion on **privacy issues in EDM**.
- Highlight **data heterogeneity challenge** in student prediction.

Paper 3

https://d.docs.live.net/ec55ae27f4c6f109/Desktop/EDM%20PROJECT/Paper3.pdf

Key Takeaways

- Used **UCI Student Performance dataset** (1044 students, 43 features).
- Performed both:
  - **Regression** (predict final grade)
  - **Classification** (pass/fail)
- **Ensemble methods performed best**:
  - Gradient Boosting → Best for regression (RMSE ≈ 3.38)
  - Random Forest → Best for classification (F1 ≈ 0.886)
- Excluded previous grades (G1, G2) to make prediction more realistic.
- Used **hyperparameter tuning (RandomizedSearchCV + 5-fold CV)**.
- Applied **SHAP for model explainability**.

Most Important Predictors

- Past failures (strongest factor)
- Study time
- School absences
- Alcohol consumption
- Parent education level

What's Useful for our Research

- Use **ensemble models (RF + Gradient Boosting)**.
- Compare **regression vs classification**.
- Add **feature engineering**.
- Include **explainable AI (SHAP)**.
- Mention dataset limitation and generalizability gap.

Paper 4

https://d.docs.live.net/ec55ae27f4c6f109/Desktop/EDM%20PROJECT/Paper4.pdf

Key Takeaways

- Educational Data Mining mainly focuses on student performance and dropout prediction.
- Random Forest, SVM, Logistic Regression, and Ensemble models perform consistently well.
- Deep Learning (especially LSTM) is useful for temporal or sequential data.
- Handling imbalanced datasets (e.g., using SMOTE) significantly improves results.
- Explainable AI methods like SHAP and LIME are important for model transparency.
- Combining multiple data sources (grades, LMS logs, behavior) improves prediction accuracy.
- Data privacy and ethical concerns remain major challenges.

What Is Important for our Paper

- Clearly define the prediction problem (performance or dropout).
- Use proper preprocessing and imbalance handling techniques.
- Compare multiple models rather than using only one.
- Evaluate using metrics like Accuracy, F1-score, and AUC.
- Include explainability (e.g., SHAP) to strengthen the research contribution.
- Highlight the practical impact of your model for early intervention.

Paper 5

https://d.docs.live.net/ec55ae27f4c6f109/Desktop/EDM%20PROJECT/Paper5.pdf

Key Takeaways

- Self-perception (especially math ability and academic expectations) is one of the strongest predictors of cognitive ability.
- Parental expectations and family support show strong and robust causal effects.
- Random Forest performed best among tested ML models.
- Using multiple explainability methods (SHAP, LIME, Morris, Feature Importance) reveals different feature priorities.
- Causal testing (PSM + robustness analysis) strengthens the reliability of findings.
- Relying on a single explainability method may give incomplete or biased interpretations.

What Is Important for our Paper

- Clearly justify why you are using explainable AI, not just prediction.
- Compare at least two explainability methods to strengthen validity.
- If possible, add causal analysis (e.g., PSM) to move beyond correlation.
- Highlight psychological and family-related variables, not just demographic factors.
- Emphasize transparency and interpretability as a research contribution.
- Discuss why model interpretability matters in educational decision-making.

Paper 6

https://d.docs.live.net/ec55ae27f4c6f109/Desktop/EDM%20PROJECT/Paper6.pdf

Key Takeaways

- Model performance depends on matching feature selection with dataset characteristics.
- Information Gain + Decision Tree performed best (up to 96% accuracy).
- Chi-Square + Random Forest also showed strong results.
- Information Gain generally outperformed other feature selection methods.
- Feature selection improved accuracy and reduced redundancy.
- DE-FS achieved high accuracy (95.8%) with only 12 features.
- There is a trade-off between accuracy (RF, NN) and computation time (DT, NB faster).

Important for our EDM Paper

- Use and justify feature selection clearly.
- Compare multiple models and report accuracy, precision, recall, F1-score.
- Consider class imbalance handling.
- Highlight efficiency and interpretability along with prediction performance.

Paper 7

https://d.docs.live.net/ec55ae27f4c6f109/Desktop/EDM%20PROJECT/Paper7.pdf

Main Useful Points

- The study used Educational Data Mining (clustering + regression analysis) to analyze elective course selection and student satisfaction.
- Four distinct student segments were identified: career-focused pragmatists, content enthusiasts, process-sensitive selectors, and balanced optimizers.
- Students differ significantly in how they select courses and what drives their satisfaction.
- Regression model explained 63% of satisfaction variance, showing strong predictive power.
- Satisfaction predictors vary across segments (e.g., career-focused students value career relevance most, process-sensitive students value administrative clarity).
- A five-layer framework was proposed: data collection, analytics, personalization, interface design, and feedback system.
- Emphasis on segment-aware personalization instead of one-size-fits-all systems.
- Ethical concerns highlighted: privacy, algorithmic fairness, transparency, and maintaining student autonomy.
- Digital divide and usability issues impact first- and second-year students more.

Most Important for our EDM Research Paper

- Use clustering to identify student segments before applying prediction models.
- Combine clustering + regression for stronger analytical contribution.
- Report variance explained (R^2) to show model effectiveness.
- Focus on explainable and personalized decision-support systems.
- Highlight information quality and expectation alignment as key educational variables.
- Link analytical findings to a practical framework or implementation model.
- Include ethical and fairness considerations in educational AI systems.

Paper 8

https://d.docs.live.net/ec55ae27f4c6f109/Desktop/EDM%20PROJECT/Paper8.pdf

Key Takeaways

- Educational Data Mining (EDM) extracts meaningful patterns from large student-related datasets.
- Uses techniques like classification, regression, clustering, association rules, and Bayesian models.
- Applications include student performance prediction, student modeling, behavior detection, and personalized learning.
- Follows a structured pipeline: data collection → preprocessing → modeling → validation.
- Major challenges: handling repeated student data (non-independence) and ensuring model generalization.

What Is Important for our Paper

- Clearly define your research objective (e.g., student performance prediction).
- Explain your dataset and preprocessing steps properly.
- Justify the choice of machine learning technique.
- Use proper validation methods (e.g., cross-validation).
- Address limitations like data dependency and overfitting.
- Show practical impact (early intervention, decision support, or system improvement).

Paper 9

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Key Takeaways

Systematic review of clustering applications in EDM (1983–2016).

- Clustering mainly used for student grouping and behavior analysis.
- Educational data requires careful preprocessing before modeling.
- No single clustering algorithm dominates across contexts.
- Research is concentrated in e-learning environments.
- Several areas (motivation, timetabling, exam failure) remain underexplored.

What Is Important for our Paper

- Avoid using clustering just for segmentation.
- Justify why your chosen method fits educational data.
- Identify and target an underexplored research gap.
- Clearly explain preprocessing and feature selection.
- Validate cluster quality using proper evaluation metrics.
- Connect findings to real educational decision-making.

Paper 10

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Key Takeaways

- Focuses on applying data mining techniques to predict student academic performance.
- Uses classification algorithms to identify at-risk students early.
- Academic records and demographic features are primary predictors.
- Model comparison shows performance differences across algorithms.
- Early prediction supports timely intervention strategies.
- Highlights importance of preprocessing and feature selection.

What Is Important for our Paper

- Go beyond basic performance prediction.
- Include psychological or behavioral variables, not just grades.
- Compare multiple models instead of reporting one.
- Use strong validation (cross-validation, robustness checks).
- Add explainability, not just accuracy metrics.
- Emphasize actionable educational impact, not just prediction.

Paper 11

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Key Takeaways

- Applies machine learning algorithms to predict student performance.
- Compares multiple classifiers to evaluate predictive accuracy.
- Feature importance analysis identifies key academic indicators.
- Data preprocessing significantly affects model outcomes.
- Ensemble methods generally outperform single models.
- Focus remains on prediction rather than causal explanation.

What Is Important for our Paper

- Do not rely only on accuracy comparison.
- Include explainability techniques to justify predictions.
- Analyze stability of feature importance across models.
- Consider causal validation to strengthen findings.
- Use diverse feature categories beyond academic scores.
- Highlight methodological contribution, not just application.

Paper 12

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Key Takeaways

- Proposes an advanced machine learning framework for student performance prediction.
- Compares traditional ML and ensemble models for accuracy improvement.
- Identifies key academic and behavioral predictors using feature analysis.
- Emphasizes model optimization and hyperparameter tuning.
- Highlights scalability of algorithms for large educational datasets.
- Focuses mainly on predictive performance rather than deep interpretability.

What Is Important for our Paper

- Move beyond accuracy and include explainability analysis.
- Demonstrate robustness using multiple validation strategies.
- Include psychological or engagement-related features.
- Compare at least two model families (e.g., tree-based vs. linear).
- Add causal or intervention-oriented interpretation.
- Clearly state how findings improve educational decision-making.

Paper 13

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Key Takeaways

• Proposes an EDM + Learning Analytics framework for Metaverse-based education evaluation.

• Uses behavioral telemetry (movement, gaze, interaction frequency) as learning indicators.

• Applies k-means clustering to identify learner archetypes.

• Uses Random Forest to predict post-assessment performance.

• Interaction diversity predicts performance better than time spent.

• Immersion improves retention but increases cognitive load.

• Identifies three learner types: Deep Divers, Passive Observers, Chaotic Explorers.

What Is Important for our Paper

• Include behavioral and engagement metrics beyond exam scores.

• Combine clustering + supervised prediction for deeper insight.

• Analyze quality of interaction, not just time-on-task.

• Consider cognitive load as a mediating variable.

• Link patterns to adaptive intervention strategies.

• Emphasize real-time or actionable educational feedback.

Paper 14

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Key Takeaways

- Bibliometric + systematic review of 793 EDM and PM studies (2000–2024).
- Research evolved in three phases: growth, consolidation, and AI-driven boom.
- Strong focus on student performance prediction and personalized learning.
- ML models (DT, SVM, NN) dominate predictive applications.
- Ethical issues (bias, privacy, transparency) are increasingly emphasized.
- Structured academic data is overused; unstructured data is underexplored.
- Qualitative and contextual impacts remain insufficiently studied.

What Is Important for our Paper

- Clearly position your work within the current AI-driven boom phase.
- Include unstructured or behavioral data, not just grades.
- Address ethics, bias, and transparency explicitly.
- Show practical implementation, not just trend alignment.
- Highlight adaptability across contexts, not single-dataset validation.
- Bridge prediction with real classroom impact and intervention.

Paper 15

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Key Takeaways

- Uses 450 students' LMS grade data across nine semesters for prediction.
- Compares kNN, Random Forest, Logistic Regression, Decision Tree, and Neural Networks.
- Binary classification outperforms multi-class classification.
- Neural Networks and kNN achieve highest F1 scores (above 0.8).
- Selecting highly correlated courses improves prediction accuracy.
- Adding weakly correlated features can reduce model performance.
- Feature selection significantly impacts predictive results.

What Is Important for our Paper

- Perform strong correlation-based feature selection.
- Compare binary vs multi-class prediction performance.
- Report F1 score for imbalanced educational data.
- Analyze how adding features affects model stability.
- Include model interpretability, not just accuracy.
- Move beyond grade-only predictors to richer behavioral data.

Paper 16

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Key Takeaways

• Systematic review of 70 studies combining XR (AR/VR/MR/Metaverse) with EDM and Learning Analytics.

• Followed PRISMA framework across six major databases (ACM, ERIC, IEEE, ScienceDirect, Scopus, WoS).

• Used LDA topic modeling, Bibliometrix, and VOSviewer for scientometric analysis.

• Documents categorized into Theoretical (21.4%), Proposal (27.1%), and Experimental studies (51.5%).

• Majority published as conference papers (58.6%), showing field is still emerging.

• Strong annual growth rate (25.35%) indicates rapidly expanding research area.

• XR + EDM enables real-time tracking of behavior, emotions, cognitive and affective states.

• Supports adaptive learning, personalization, collaboration, and engagement.

• Improves self-efficacy, self-regulated learning, retention, and academic achievement.

What Is Important for our Paper

• If doing a review, clearly justify database selection and keyword query strategy.

• Use strict inclusion/exclusion criteria and report screening numbers transparently.

• Categorize studies logically (theoretical vs proposal vs experimental).

• Combine qualitative content analysis with quantitative bibliometric methods.

• Show measurable impact (learning gains, engagement, retention), not hype about XR.

• Critically assess maturity of the field — avoid treating conference-heavy evidence as strong validation.

• Identify research gaps instead of repeating that “XR improves learning.”